An Overview of Convertigo

Convertigo is an open source mobile app development platform that enables enterprises to integrate any mobile application with back-end applications or data sources — all at a considerably lower cost and within less time.

Convertigo is the first open source mobile platform to provide a complete, end-to-end solution — from back-end enablement to mobile UI development tools — integrated in one unique mobile application development platform (MADP) and MBaaS (Mobile Backend-as-a-Service).

The platform comprises several components that include the Convertigo server, Studio and third party SDKs. Convertigo delivers a secured and scalable, all-in-one solution that integrates rapid cross-platform mobile development tools and a powerful MBaaS. It provides challenging back-end enablement and features middleware optimised for mobility.

Architecture of the mobile platform

A mobile platform differs from a simple mobile application development tool by providing all the components needed to build, run, manage and connect mobile applications to the existing enterprise information system.

The capabilities that a mobile platform should possess are listed below.

- *Mobile back-end connectors:* These allow mobile apps to connect to the enterprise database and to business applications.
- *Mobile service orchestrator:* This allows back-end data to be aggregated, filtered and combined to provide a mobile-friendly service API. The orchestrator can also augment an existing back-end application with mobile-specific capabilities such as push notifications or locator services.
- *Cross-platform UI:* This allows developers to work on multiple operating systems.
- *Security manager:* This is used to encrypt sensitive data on the network or on the mobile device.
- *Mobile application SDKs:* These provide the capabilities to integrate other third party mobile UI development efforts.

What are mobile services?

Mobile applications need mobile services to interact with the data. Mobile services are made on top of existing back-end services provided by ESBs (enterprise service bus) or other SOA based architecture. Mobile applications interact with mobile services using standard protocols like HTTP/HTTPS, XML format, etc.

Mobile services can be defined using either a bottom-to-top approach or a top-down approach, where the service model is defined by the mobile UI developer. A very